Integrating Metabolic Mathematics, Behavioral UI/UX, and Predictive Recommendation Systems to Mitigate Mobile Health Application Abandonment

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***Abstract*—Retention rates for mobile health, fitness, or any self-help applications, despite being mature in resources and progression, lack significant long-term retention among their users. This paper investigates the integration of metabolic mathematics, behavioral User Interface (UI) and User Experience (UX) design architectures, and advanced filters to create recommendation systems to cater toward users' needs, in order to solve the many friction points driving self-help application abandonment [9], [10]. When analyzing intersection points of physiological tracking (i.e., Mifflin-St Jeor, Kevin Hall's dynamic weight models) [1], [12] close to behavioral frameworks (i.e., BJ Fogg's Behavior Model) [11] and neural collaborative filtering [7], [8], we end up with a framework that is clean and unified to keep retention high. This paper demonstrates that optimizing the "Time to First Value" (TTFV) through cognitive-load reduction and personalized, adaptive feedback loops reduces 30-day retention fall-off for self-help applications [9], [10], [13].**

***Index Terms*—Behavioral UI/UX, Mobile Health Retention, Neural Collaborative Filtering, Metabolic Mathematics, Cognitive Friction**

I. Introduction

Rapid global scaling of self-help applications democratized access to wellness management. However, in industry practice, approximately 25% of self-help applications are abandoned after one interaction, and median retention rates by users drop below 10% by day 30 [9], [10]. The withering retention rates provide a gap between engagement and retaining knowledge, most of the time before the user can even form a habit, which empirical research shows requires a median of 59 to 66 days [13].

Traditional programs treat diet management and physical training as static and compliance-driven [1]–[6], yet human behavioral modification is actually heavily dynamic and vulnerable to friction [10], [11]. Sustainable programs must adjust to the user by blending these three pillars:

- **Quantitative Physiology:** Getting quantitative data from human energetics and metabolism for predictive models [2], [12].
- **Predictive Analytics:** Using machine learning architectures for personalization and exercise trajectories without needing to over-survey the user [7], [8].
- **Behavioral Design:** Lower cognitive barriers, build intrinsic motivation, and procure positive habit loops on the UI/UX [11], [13].

This paper explores how these pillars converge from contextualizing metabolic science to fit the boundaries of modern recommendation algorithms & cognitive physiology. We map out a blueprint for the future of wellness apps with high-retention, habit building, engaging, and effectiveness.

II. Literature Review

A. Diet Management & Metabolic Math

Obeying some sort of diet is the primary driver of success when it comes to body composition and chronic disease management, such as Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) [1]–[3]. However, effective prescription heavily relies on the underlying mathematical formulas used to estimate a baseline for a human [12]. Historically, the Harris-Benedict Equation served to be the standard for calculating Basal Metabolic Rate (BMR):

$$\text{BMR}_{\text{male}} = 66.4730 + 13.7516W + 5.0033H - 6.7550A \quad (1)$$

$$\text{BMR}_{\text{female}} = 655.0955 + 9.5634W + 1.8496H - 4.6756A \quad (2)$$

where W = weight in kg, H = height in cm, and A = age in years.

Many digital platforms prefer the Mifflin-St. Jeor Equation due to being more accurate in modern populations [1], [2]:

$$\text{BMR} = 10W + 6.25H - 5A + s \quad (3)$$

where $s = +5$ for males and -161 for females.

Despite being useful for finding BMR, it fails to account for metabolic adaptation, or how energy depletes during prolonged caloric deficits [12]. Dr. Kevin Hall's thermodynamic modeling (NIH Body Weight Simulator) demonstrates that body mass alteration is a dynamic system driven by shifting macronutrient oxidation fluxes and changes in lean mass versus adipose tissues [12]. Therefore, static caloric tracking apps can frustrate the user when linear weight loss plateaus occur, leading straight to app abandonment [10], [12].

B. Exercise Progression & Physical Conditioning

Physical conditioning requires keeping a balance of injury mitigation with progressive overload, which means to gradually increase volume, frequency, or intensity [4]–[6]. According to guidelines set by the National Strength & Conditioning

Association (NSCA) and the American Council on Exercise (IFT Model), training protocols must map to distinct user experience thresholds [5], [6]:

TABLE I
USER CLASSIFICATION AND FITNESS PROGRESSION METRICS

Classification	Resistance Variable	Cardio Strategy	Core Objective
Beginner	Form acquisition, low intensity	Low-impact steady-state	Neuromuscular adaptation [5], [13]
Intermediate	Step-loading, 3–4 sets	Anaerobic thresholds	Hypertrophy & endurance [6]
Advanced	Autoregulated periodization	High-intensity interval	Maximize athletic power [4], [6]

When digital applications fail to adapt to a user's changing fitness level such as over-intensifying to a beginner or failing to challenge someone who's advanced, the imbalance causes the user to have anxiety with injury or boredom. These imbalances are another reason a user would be churned away from the app [10].

C. Recommendation Algorithms in Personal Health

To keep users engaged, digital platforms must serve relevant content (recipes, workouts, and educational blocks) without overwhelming them [7], [8]. Modern recommendation architectures have evolved beyond basic popularity metrics:

- **Collaborative Filtering (CF):** Historical behavior is computed on a vector (i.e., cosine similarity & Pearson correlation). If User A and User B share identical recipe tracking logs, the system recommends User A's unique selections to User B [8].
- **Matrix Factorization (MF):** Sparse user-item interaction matrices are decomposed into a lower dimension factor, and the user base's preferences are captured [7].
- **Neural Collaborative Filtering (NCF):** Linear matrix operations are replaced with neural networking, allowing the model to learn complex, non-linear relationships between traits of the user and of their wellness [7], [8].

All three of these models can predict struggling points of beginners and more accurately predict someone's diet from craving patterns [7]. Thus, these models would also proactively solve adaptation problems and keep a pattern of recommendation that is not overwhelming [8], [10].

III. METHODOLOGY

To evaluate the factors that drive an adoption of a wellness app rather than abandonment, we will analyze them through BJ Fogg's Behavior Model ($B = MAP$) [11]. The model states behavior happens when Motivation (M), Ability (A), and Prompt (P) converge:

$$B = M \cdot A \cdot P \tag{4}$$

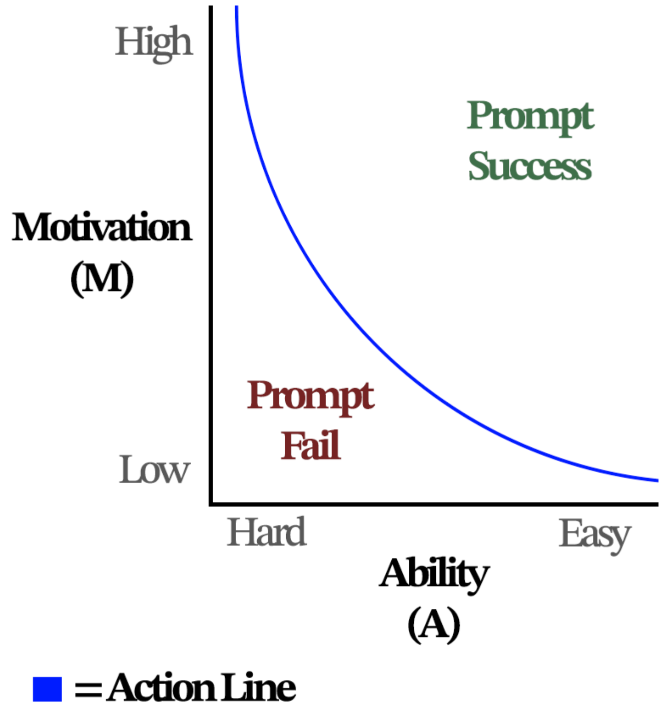

Fig. 1. BJ Fogg's Behavior Model ($B = MAP$), illustrating the activation threshold relationship between Motivation and Ability.

In this study, Ability is defined as the ease of User Experience (UX) [11], which could mean speed to log a meal, change a setting, or starting a workflow, while Motivation is driven by biological feedback [2], [12], which gamification can help drive.

With the benchmark in mind, apps have high abandonment because of manual entry, linearity, lack of personalization, lack of gamification, and excessive notifications which poorly prompt the user to take action [9], [10]. Apps that have low abandonment track and automate for the user, only leaving them to take action toward something that is easily digestible and requires little cognitive load [10], [11]. Then, the model builds off the user's habits over time [13], giving more personalized routes to their wellness journey [7], [8].

IV. ANALYSIS

A. Behavioral UI/UX Failures That Kill Retention

Analyzing high-abandonment platforms will show that abandonment is not often caused by the accuracy of the workout or diet plan [10]. One main cause of it is cognitive friction [11].

Cognitive friction is a situation where, for example, the meal logging UI requires the user to manually input ingredients, then search through databases that aren't indexed, and manually calculate portion weights. These drive their cost of action, and it quickly depletes their motivation [11]. That dynamic alone can cause abandonment on the first day of downloading the app [9], [10]. Apps that combat the friction design their app around Time to First Value (TTFV), showing the user they're making progress early in the habit-formation cycle [10], [13]. The app later gets familiar with the user's history through recommendation models [7], [8] and shortens the TTFV; that scenario drives a parallel that's similar to ordering "the usual" at a local place you visit often.

B. *Unified Architecture*

To achieve high, long-term retention, the app needs to apply a certain feedback loop. First, the user inputs biological data, and the app will recommend a starting point using standard baselines [1], [2], predicting the next step from there. Secondly, through the behavioral interface, it would reduce cognitive friction through predictive elements [11]. Lastly, the app will automatically update workout progression [5], [6] and adjust caloric deficits (for plateaus) from NCF input [7], [12].

Then, when an intermediate user hits a dynamic physiological weight plateau, predicted by Kevin Hall's metabolic math [12], the NCF system flags the trend [7]. Usually, the user would abandon the app from there after seeing the plateau [10], but instead, the model proactively adjusts the interface so there isn't a dramatic shift in progress that throws off the user's balance. It will alter the caloric target or serve an automated progressive overload training adjustment [5], [12]. The small nudge in behavior will maintain user alignment through following Fogg's action curve [11]. The experience is still challenging the user, but it doesn't push them past their capability limits [6], sustaining interaction throughout the $\sim$60-day window needed for long-term habit formation [13].

V. CONCLUSION

Longevity of digital health apps depends on a balance between psychology and medical science. High app abandonment is caused by psychology and user experience failures [9]–[11], no matter if the app creator knows of the user intent or the metabolic science to make their lives better [1]–[6], [12]. Apps have to ground their systems with adaptation, recommendation, and low cognitive friction [7], [8], [11]. Future research should continue to explore automation and non-invasive tracking (whether it be something like a wearable watch to track biometrics) to further curb friction and drive long-term retention, and more importantly, habit formation [13].

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